Lecture 15 – Multi-objective Particle Swarm Optimisation

ECM3412/ECMM409 – Nature-Inspired Computation

Dr David Walker

Department of Computer Science
D.J.Walker2@exeter.ac.uk

Autumn 2025

Basic PSO

Particle Swarm Optimisation

1. Create and initialise N random particles on the search space
2. For each timestep:
    1. For each individual:
        1. Update the position of the particle by adding a velocity to the current position of the particle
        2. Update the velocity

$$x_{ij}(t+1) = x_{ij}(t) + v_{ij}(t+1)$$

$$v_{ij}(t+1) = \mathbf{v_{ij}(t)} + \mathbf{c_1} \times \mathbf{z_1} \times (\mathbf{p_{ij}} - \mathbf{x_{ij}}) + \mathbf{c_2} \times \mathbf{z_2} \times (\mathbf{g_j} - \mathbf{x_{ij}}),$$

where

- $\mathbf{v_{ij}(t)}$ is the velocity at time t
- $\mathbf{c_1} \times \mathbf{z_1} \times (\mathbf{p_{ij}} - \mathbf{x_{ij}})$ is personal information
- $\mathbf{c_2} \times \mathbf{z_2} \times (\mathbf{g_j} - \mathbf{x_{ij}})$ is global information

Introduction

Use PSO for multi-objective problems

- Need to identify **leaders** according to multiple objectives

Multi-objective **pbest**

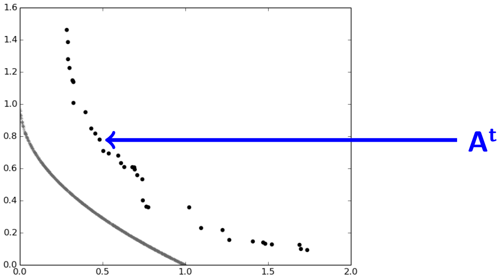
Assuming
$\mathbf{y}_i = (f_1(\mathbf{x}_i), \dots f_M(\mathbf{x}_i))$, and
$\mathbf{p}_i$ is the corresponding **pbest** for particle i

- Only update if $\mathbf{y}_i$ dominates $\mathbf{p}_i$ — **keep the oldest**
- Always update unless $\mathbf{p}_i$ dominates $\mathbf{y}_i$ — **keep the newest**

Archives

The **archive** A^t represents the Pareto front approximation at time t

- All members are **mutually non-dominated** – no dominated solutions
- New solutions in the swarm P_t and the archive A^t must be compared in terms of dominance to update solutions in the archive
  - Newly dominated solutions are removed
  - If the solutions in P_t are not dominated by the archive then they are added to A^t



Updating the Archive

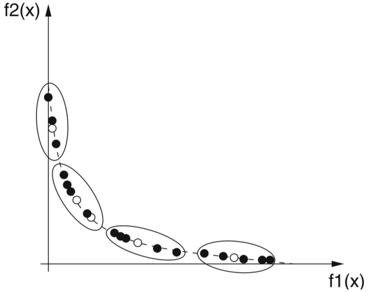
Given a new solution $\mathbf{y}$

1. For each solution $\mathbf{a}_i$ in the archive $\mathbf{A}^t$
   1. If $\mathbf{y}$ dominates $\mathbf{a}_i$ then remove $\mathbf{a}_i$
   2. If $\mathbf{a_i}$ dominates $\mathbf{y}$ then proceed to the next member of $\mathbf{A}^t$ and mark $\mathbf{y}$ as **dominated**
2. If $\mathbf{y}$ is not marked as dominated add it to $\mathbf{A}^t$

Limiting the size of the archive

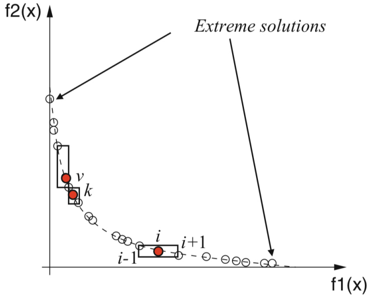
- Often $\mathbf{A}^t$ is **limited** in size – there is an upper limit on the number of solutions it can contain
- Computationally expensive to find leaders in a large archive
- Not useful for a decision maker to be presented with a large number of solutions
- Use a **diversity preserving** mechanism to ensure a diverse set of solutions is maintained

Archive diversity – clustering



- Use **clustering** to identify a representative subset – maintaining the characteristics of the original subset
- Zitzler (1999) used **hierarchical clustering** to do this
- Join adjacent clusters until the required number of clusters is reached
- At the end keep the solution with the minumum average distance to the cluster center

Archive diversity – crowding distance



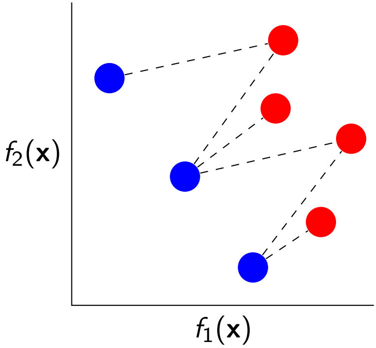
- The same approach used to promote diversity in NSGA-II (Deb *et al.*, 2002)
- Score solutions according to their proximity to their nearest neighbours – those in crowded areas receive low scores and those in isolated areas receive high scores
- Score of ∞ for extremes to ensure they are retained

Leaders – random selection

- In a multi-objective context there is no single **global best** – we want to attract the swarm toward the set of non-dominated solutions lying on the Pareto front
- The non-dominated solutions stored in the archive are a good source of leaders
- A basic approach is to randomly select members of the archive
- Alternatively **maintain diversity** by preferring solutions in areas of the archive that are sparser

Leaders – dominated solution count

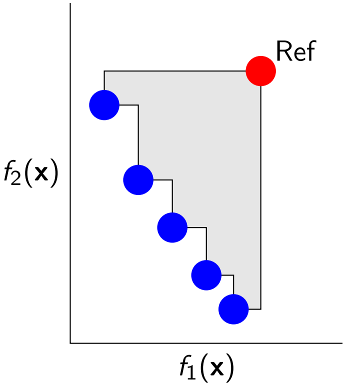
- Rank the solutions in archive based on the number of solutions in the swarm that are dominated by them
- Archived solutions that dominate a larger number of solutions in the swarm are less likely to be selected



Leaders – hypervolume

Hypervolume

The area or volume dominated by the non-dominated solutions between the non-dominated solutions and a reference point

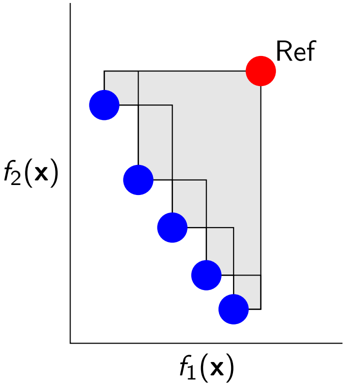


- Use the solution for which a leader is required as the reference point
- The archive member whose contribution to the hypervolume is maximum is the leader
- If the solution is not dominated by the archive members (and there is no reference point) then one is chosen at random

Leaders – hypervolume

Hypervolume

The area or volume dominated by the non-dominated solutions between the non-dominated solutions and a reference point

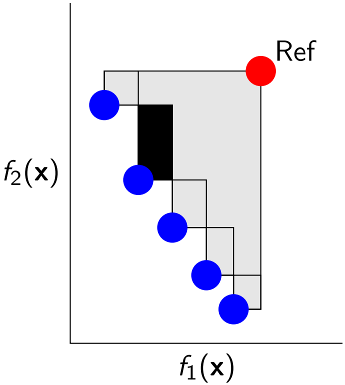


- Use the solution for which a leader is required as the reference point
- The archive member whose contribution to the hypervolume is maximum is the leader
- If the solution is not dominated by the archive members (and there is no reference point) then one is chosen at random

Leaders – hypervolume

Hypervolume

The area or volume dominated by the non-dominated solutions between the non-dominated solutions and a reference point



- Use the solution for which a leader is required as the reference point
- The archive member whose contribution to the hypervolume is maximum is the leader
- If the solution is not dominated by the archive members (and there is no reference point) then one is chosen at random

Many-objective PSO

Three issues with many-objective optimisation

1. Problems with **visualising** a high-dimensional Pareto front approximation
2. **Reduction in the search** capability of multi-objective optimisers
3. Exponential **increase in the number of solutions needed** to approximate the Pareto front

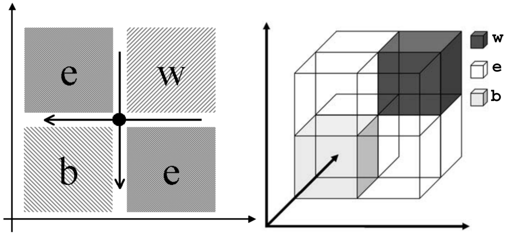


Fig. 1. Schematic view of Pareto-dominance based partial order in 2D and 3D problems when a candidate solution is considered (•): equal (e), better (b) and worse (w) solution regions are shown.

H. Ishibuchi, N. Tsukamoto & Y. Nojima, "Evolutionary Many-objective Optimisation: A Short Review", Proc. IEEE Congress on Evolutionary Computation, 2008

M. Farina & P. Amato, "Fuzzy Optimality and Evolutionary Multiobjective Optimization", Proc. Int. Conference on Evolutionary Multi-criterion Optimisation, 2003

Average rank

Rank the archived solutions M times – once per objective – and then take the average of these ranks

- r_{im} is the rank of the i-th individual according to the m-th objective
- Average rank is then

$$\bar{r}_{im} = \frac{1}{M} \sum_{m=1}^{M} r_{im}$$

f_1	f_2	f_3	f_4	f_5	r_{*1}	r_{*2}	r_{*3}	r_{*4}	r_{*5}	$\bar{r}$
0.47	0.80	0.62	0.73	0.89	2	4	4	4	4	3.6
0.07	0.40	0.36	0.66	0.24	1	2	3	3	1	1.8
0.63	0.48	0.01	0.01	0.55	3	3	1	1	3	2.2
0.78	0.13	0.37	0.09	0.28	4	1	2	2	2	2.4

- Individuals with better (usually **lower**) ranks are more likely to be selected as leaders

Distance ranking

Compute the rank r of solution $\mathbf{x_i}$ with

$$r(\mathbf{x_i}) = \sum_{m=1}^{M} \sum_{j=1, j \neq i}^{N} d_{ijm},$$

where

$$\mathbf{d}_{ij} = (|f_1(\mathbf{x}_i) - f_1(\mathbf{x}_j)|, \ldots, |f_M(\mathbf{x}_i) - f_M(\mathbf{x}_j)|)$$

- $\mathbf{d}_{ij}$ is the absolute distance between each objective values of the two solutions $\mathbf{x}_i$ and $\mathbf{x}_j$
- $r(\mathbf{x}_i)$ prefers solutions that are in uncrowded regions of the space

Summary

Multi-objective PSO

- As with GAs the basic algorithm can be used but modifications must be made to accommodate comparisons based on multiple objectives
- Can again use dominance for **multi-objective** problems
- An alternative is needed for **many-objective** problems – average rank, distance-based, and others

Next time

- Nature-inspired machine learning